

01 – Executive Story (Bind Compute)

1. What Bind is building
A developmental computing stack where **certified computational matter**—discovered under causal tes

2. The problem
Today's "AI on chips" usually means a fixed model running on GPUs/accelerators. Changing capability

3. The core insight
Treat computation as **matter that must earn its place**: discover → intervene/certify → archive wit

4. What is different

- Causal certification before persistence
- Single archive as construction material
- Embodied discovery path (ECR) that can feed the foundry
- Faithful execution semantics (sum+tanh IR)
- Generic RCF host: **one bitstream**, many organizations via config

5. What has actually been built
Software: Digital Chemistry substrate, Machine Factory, archive, CLM/PCM, ECR, science spine S1-S4,
Hardware path: faithful IR, RTL, **generic multi-slot RCF iCE40 bitstream** (135100 bytes).
Live board execution and physical robot: **pending**.

6. Flagship demonstration
Object `ECR2MF\_g0\_field\_s1002`: unknown-world discovery → certificate → archive → x211 reuse → neces
Runnable without the full IP tree: `STANDALONE\_FLAGSHIP/`.

7. Architecture (one line)
World → ECR → certify → MF/archive → matter package → RCF → body → world.

8. Why hardware matters
Without a reconfigurable host, each organization is a dead-end netlist. RCF makes organization a **l**

9. Roadmap
Board hot-load → minimal physical body → reopen developmental edge (robot→ECR→MF→RCF) → multi-node f

10. Why this could become a new architecture
Not larger neural nets—**developmental continuity of certified organization** from discovery to sili